RAG Assignment

Topic: Baseline Retrieval vs. Cross-Encoder Re-Ranking on the GDPR
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Course: Large Multimodal Models and Applied NLP
Document Analysed: Regulation (EU) 2016/679 (GDPR, 88 pages, 405 chunks)
Models Used: bge-small-en-v1.5 (Bi-Encoder), ms-marco-MiniLM-L-6-v2 (Cross-Encoder), qwen/qwen3.8-27b (via Groq API)
Repository: https://github.com/GauravPatel369/LMA_A1

1. Architectural Comparison: Bi-Encoder vs. Cross-Encoder

A **bi-encoder** independently embeds queries and documents, enabling fast similarity search using FAISS. In my experiments, query embedding took 11.60 ms and vector search took 0.07 ms. However, semantic similarity can miss legal nuances, negations, and exceptions.

A **cross-encoder** jointly processes the query and passage using cross-attention, enabling more precise relevance scoring. However, it requires a separate forward pass for each query-passage pair, making it more computationally expensive.

Therefore, the bi-encoder is suitable for initial retrieval, while the cross-encoder is useful for re-ranking a small candidate set.

2. Dataset Processing and Experimental Setup

I used the official GDPR text, containing 173 recitals and 99 articles. During preprocessing, I removed repeated page headers and footnote citations to avoid false keyword matches.

A structure-aware chunker used a 220-word window with 40-word overlap, respecting recital and article boundaries. This produced 405 chunks, with notebook self-tests verifying the extraction of all 173 recitals and 99 articles.

Both pipelines used the same dataset and generation prompt:

- **Baseline:** BGE embeddings -> FAISS top 10 -> top 3 passages.
- **Re-ranked:** BGE embeddings -> FAISS top 10 -> cross-encoder scoring -> top 3 passages.
- **Generation:** Qwen3.8-27B via Groq, using temperature 0.

This setup isolates the effect of re-ranking on the retrieved context.

3. Rank Shifts and Answer Comparisons

Across my 14 test queries (12 failure probes and 2 control queries), Recall at 3 increased from 7/14 (50%) in the baseline to 9/14 (64.3%) after re-ranking. The Stage 1 ceiling (Recall at 10) was 12/14 (85.7%).

Query Rank Shift Summary:

Query	Target Provision	Baseline Rank	Re-Ranked Rank	Net Result
Q1 Automated loan refusal (negation/jargon)	Recital 71	Rank 4	Rank 1	Rescued into top 3
Q2 Tell individuals about breach (keyword trap)	Article 34	Rank 5	Rank 2	Rescued into top 3
Q3 Two companies decide data use	Article 26	Rank 8	Rank 1	Rescued into top 3

jointly				
Q4 Privacy built into design	Article 25	Rank 4	Rank 7	Remained outside top 3
Q5 Contract terms with data processor	Article 28	Rank 6	Rank 7	Remained outside top 3
Q6 Duration allowed to keep records	Article 5(1)(e)	Rank 1	Rank 4	Regressed out of top 3
Q7 Technical measures against database theft	Article 32	Not in top 10	Not in top 10	Missed by Stage 1
Q8 Foreign court demanding European data	Article 48	Not in top 10	Not in top 10	Missed by Stage 1
C1 Maximum administrative fine (Control)	Article 83	Rank 1	Rank 1	Retained at Rank 1
C2 Definition of personal data (Control)	Article 4	Rank 1	Rank 1	Retained at Rank 1

Out of the 5 queries with headroom (where the gold passage sat between ranks 4 and 10), re-ranking brought 3 into the top 3 (Q1, Q2, and Q3).

Q1 - Negation and legal terminology: Re-ranking moved Recital 71 from rank 4 to rank 1. The baseline lacked the relevant provision, whereas the re-ranked context enabled the generator to explain the restrictions and exceptions concerning solely automated decisions.

Q2 - Keyword trap: The baseline retrieved Article 33, concerning notification to supervisory authorities, instead of the more relevant Article 34. Re-ranking moved Article 34 from rank 5 to rank 2, enabling the answer to include the requirements and exceptions for notifying affected individuals.

Q3 - Buried legal provision: Article 26 moved from rank 8 to rank 1, allowing the answer to explain the agreement required between joint controllers.

Q6 - Re-ranking regression: Article 5(1)(e) fell from rank 1 to rank 4. The cross-encoder preferred descriptive recitals containing related retention terminology over the concise statutory provision, weakening the generated answer

Retrieval ceiling: Articles 32 and 48 were absent from the initial top 10. Re-ranking could not recover them because it only scores retrieved candidates

Overall, re-ranking improved top-3 recall but did not consistently improve every query.

4. Latency Profiling

Median latency was measured across 10 runs, excluding the first cold-start run.

Stage Latency Summary:

Stage	Median Latency
Stage 1: Query embedding (bge-small-en-v1.5)	11.60 ms

Stage 1: FAISS vector search (top 10 over 405 chunks)	0.07 ms
Pipeline 1 Total (Baseline Retrieval)	11.67 ms
Stage 2: Cross-encoder re-ranking (10 candidate pairs)	23.77 ms
(Average cost per candidate pair)	2.38 ms / pair
Pipeline 2 Total (Two-Stage Retrieval)	35.44 ms
Generation: LLM Answer 1 (Groq API)	452.62 ms
Generation: LLM Answer 2 (Groq API)	640.72 ms

Re-ranking increased retrieval time by roughly three times (from 11.67 ms to 35.44 ms). However, an extra 24 ms is small when compared to the 450 to 640 ms taken by the LLM to generate text (roughly 5% of the total turnaround time). In practice, adding 24 ms to improve recall from 50% to 64% and retrieve missing legal exemptions is an acceptable trade-off.

5. Scaling: Why Not Cross-Encode the Entire Corpus?

One natural question is whether we can skip the bi-encoder entirely and score all passages with the cross-encoder from the start.

This comes down to computational scaling:

- A bi-encoder's work is pre-computed: Generating embeddings for all 405 chunks took 1.85 seconds once during indexing. At search time, only the user query is embedded (11.6 ms), and vector search scales logarithmically or in constant time.
- A cross-encoder cannot pre-compute document representations: Because the model requires both the query and the chunk tokens together, no document vector exists beforehand. Every search requires N full forward passes through the transformer, meaning computation scales linearly with the number of chunks.

I measured this in Section 14 of my notebook by scoring a probe query against all 405 chunks:

- Scoring all 405 chunks took 0.634 seconds (about 1.56 ms per pair).
- Re-ranking the top 10 candidates took 0.024 seconds (23.8 ms).
- Full-corpus cross-encoding was already 27 times slower on just 405 chunks.

Extrapolating this cost of 1.56 ms per pair shows why it does not scale:

- 10,000 chunks (about 50 policy documents): 1.56 ms x 10,000 = 15.6 seconds per query.
- 100,000 chunks (an enterprise document base): 1.56 ms x 100,000 = about 2.6 minutes per query.
- 1,000,000 chunks (a large legal database): 1.56 ms x 1,000,000 = about 26 minutes per query.

This confirms why two-stage retrieval is standard practice: the bi-encoder quickly narrows the search space to 10 candidates in milliseconds, and the cross-encoder spends its computation only on that shortlist.

6. Closing Observations

From these experiments, I took away three main practical points:

1. Re-ranking is valuable when queries involve negations or everyday phrasing, where vector similarity tends to leave important clauses between ranks 4 and 10.

2. The initial candidate window should be widened to 30 or 50 chunks, ideally combining dense retrieval with BM25 keyword search, so that the pipeline is not limited by the Stage 1 recall ceiling seen in Q7 and Q8.

3. System builders should watch for regressions like Query 6, where wordy recitals outscore concise statutory rules. Adding a lexical match or article booster can help keep primary principles at the top.